

Q.2 a) We have: $\frac{dy}{dt} = 2ty - t^3y$, $0 \leq t \leq 2$, $y(0) = 1$

$$\Rightarrow \frac{dy}{y} = (2t - t^3)dt \Rightarrow \ln|y| = t^2 - \frac{t^4}{4} + C$$

$$\text{at } t=0, y(0)=1 \Rightarrow C=0$$

$$\Rightarrow \ln|y| = t^2 - \frac{t^4}{4} \Rightarrow \boxed{y = \exp\left(t^2 - \frac{t^4}{4}\right)}$$
 Analytical Solution

So in our code, we simply draw the graph:

$y = e^{t^2 - t^4/4}$ from $0 \leq t \leq 2$, one .py file for this.
 $\Rightarrow$ Then, observe the analysis.py file

b) Some preliminary rough work:

$$\frac{dy}{dt} = 2ty - t^3y$$

$\Rightarrow$ Forward Euler: (Explicit Euler)

$$\text{update: } \frac{y^{n+1} - y^n}{\Delta t} = 2ty^n - t^3y^n$$

$$\Rightarrow y^{n+1} = y^n + y^n \cdot \Delta t (2t - t^3) = \boxed{y^n [1 + \Delta t (2t - t^3)]}$$

all values given $\checkmark$

$\Rightarrow$ incrementally construct vector values

$\Rightarrow$ Backward Euler: (Implicit Euler)

$$\frac{y^{n+1} - y^n}{\Delta t} = 2ty^{n+1} - t^3y^{n+1} \Rightarrow y^{n+1} \left[\frac{1}{\Delta t} - 2t + t^3 \right] = \frac{y^n}{\Delta t}$$

$$\Rightarrow \boxed{y^{n+1} = \frac{y^n}{1 - \Delta t (2t - t^3)}}$$

* Keeping this in mind, will make 2 files [linespace & 3 schemes] and obtain 15 outputs

$\Rightarrow$ Final job: to graph all values in .py file, save as png.

* Acknowledgement: Main.cpp [.txt file uploading, obtaining .py file: results array, used AI]

Other two .cpp files: AI gave incorrect code, made multiple tweaks